

# Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C<sub>60</sub> Graphullerene

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(Dated: June 19, 2026)

Quasi-hexagonal (qHP) C<sub>60</sub> graphullerene hosts polaronic charge transport in which nearest-neighbor coupling  $J$ —not band dispersion—sets mobility scales [1]. Here we quantify how *joint* B/N/P substitution and biaxial strain reshape the *total-energy* landscape of finite qHP supercells, independent of a full Marcus transport closure. PBE+DFT-D3 calculations on tetramers (36 converged neutral points) and periodic  $n \times \text{C}_{60}$  cells ( $n = 1\text{--}8$ ; 40 converged points) yield dopant-resolved  $\pi$ -gap evolution, opposite-sign mechanical coefficients  $\alpha$  for N ( $-303$  meV/%) vs. B ( $+57$  meV/%), and a synergy order parameter  $\mathcal{S}$  [Eq. (1)] with  $|\mathcal{S}|$  up to  $\sim 32$  meV/atom at  $+3\%$  strain. Across  $n = 1\text{--}8$  and B/N/P dopants (15 coupled points at  $+3\%$ ), the effects of simultaneous strain and substitution are *not* additive when applied sequentially. At tetramer scale, a  $2 \times 2$  factorial IPR/ $J$  analysis (pristine/B-doped  $\times \epsilon=0/+3\%$ ) reveals that B substitution dominates delocalization ( $\Delta J \approx +20$  meV at  $\epsilon=0$ ), while  $+3\%$  tensile strain reduces  $J$  under B doping ( $46 \rightarrow 37$  meV); all four points remain in the small-polaron regime ( $J \ll \lambda/2$ ).

## INTRODUCTION

Synthetic carbon allotropes with extended  $\pi$  conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [1, 2]. The quasi-hexagonal phase (qHP C<sub>60</sub>, graphullerene) [2] covalently links C<sub>60</sub> units, shortening interfullerene distances by  $\sim 0.9$  Å and raising mobilities by nearly two orders of magnitude over vdW C<sub>60</sub> [1]. Capobianco *et al.* [1] showed that the gain tracks enhanced nearest-neighbor coupling  $J$  ( $\sim 75\text{--}81$  meV vs.  $33$  meV) within a hopping model, with similar polaron binding ( $\lambda^- \sim 0.1$  eV) in both phases.

Beyond pristine monolayers, strain and substitution independently tune qHP C<sub>60</sub> band edges [3–5]. Non-additive strain–doping coupling has been reported in other two-dimensional networks [6]; a quantitative energy cross term  $\mathcal{S}$  [Eq. (1)] has not been reported for substitutional B/N/P qHP C<sub>60</sub> tetramers and  $n = 1\text{--}8$  supercells under a common PBE+D3 protocol. If formation energies or strain penalties are estimated from *separate* doping-only and strain-only scans, the omitted cross term can reach tens of meV/atom in coupled cells (this work)—large enough to mis-rank dopant stability or strain set-points in screening workflows.

We combine Capobianco-style observables (gaps, IPR,  $J$ ) with systematic strain–doping energy scans on shared tetramer and size-scaling grids. The present study establishes the *energy* coupling metric  $\mathcal{S}$  and band-edge trends; Marcus  $\lambda$  and rate estimates are treated in Supplemental Material and are not required for the non-additivity conclusions here.

## COMPUTATIONAL METHODS

### Electronic-structure setup

All total-energy and electronic-structure calculations use CP2K [7] with PBE [8] and Grimme DFT-D3 (Becke–Johnson damping) [9], as documented in Supplemental Material. This level is deliberately chosen for a uniform, size-converged scan of  $(\epsilon, \delta)$ ; it differs from the Koopmans-compliant PBE( $\alpha_K$ )+rVV10 framework of Ref. 1, which targets absolute polaron binding and mobilities in pristine supercells.

**Rigid-strain approximation.** Unless stated otherwise, biaxial strain is imposed by scaling the in-plane lattice vectors of the *pristine* equilibrium cell at *fixed fractional atomic coordinates* (no atomic relaxation under load), not by re-scaling each dopant-relaxed equilibrium lattice. Reported  $\alpha$  therefore mixes dopant-induced changes in strain response with any equilibrium lattice offset relative to pristine. This protocol upper-bounds the strain energy and is applied consistently across the comparative grids below. In related 2D carbon networks, allowing ionic relaxation under load typically reduces strain-energy derivatives relative to rigid cells [3]; we therefore interpret reported  $\alpha$  and  $\mathcal{S}$  as *upper bounds* on the rigid-lattice coupling, with relaxed-ion values expected to be smaller in magnitude but not necessarily vanishing.

Plane-wave cutoff is 300 Ry for cells with  $n \leq 5$  fullerene units and 280 Ry for  $n \geq 6$  (memory limit for  $6 \times 8$  cells on our hardware). Because  $\mathcal{S}(n)$  compares energies *within* each  $n$  at identical cutoff, trends at fixed  $n$  are internally consistent. Cross- $n$  comparisons—including the N  $\mathcal{S}$  sign change between  $n = 6$  and  $n = 8$ —coincide with the 300→280 Ry switch and should be interpreted cautiously until a matched-cutoff check at

$n = 6$  is available (Supplemental Material Table S2).

SCF convergence criterion  $10^{-6}$  Eh with OT minimization; basis DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

### Substitutional doping protocol

Tetramer models ( $4 \times C_{60}$ , 240 C sites) use substitutional B, N, or P at 5% of carbon sites ( $\sim 12$  replacements per cell), chosen by a reproducible pseudo-random protocol (fixed seed 42) over all C coordinates—not a symmetry-averaged or thermodynamic minimum. Periodic size-scaling cells place **one** substitutional dopant per  $C_{60}$  unit (fixed count, so nominal concentration  $\sim 1/60$  per molecule decreases as  $n$  increases). The two protocols differ in *nominal dopant concentration* (5% vs.  $\sim 1.7\%$  at  $n = 4$ ); tetramer  $\alpha$ , gaps, and IPR/ $J$  are therefore trend indicators, not directly interchangeable with  $n=4$  periodic energies without a matched-concentration benchmark. We do not re-optimize dopant sites; all reported  $\alpha$  and  $\mathcal{S}$  refer to these fixed, reproducible configurations and should be read as site-specific trends pending tests on alternate placements or thermodynamic minima.

### Computational grids

In total,  $\gtrsim 90$  converged CP2K tasks support the main-text claims (Table S2); the principal neutral-energy grids are:

**Mechanical strain response (24 calculations).** Pristine and B/N/P tetramers at  $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$  supply total energies for  $\alpha = dE/d\epsilon$  and substitution energies  $E_f$  (Table S1).

**Electronic structure (12 calculations).** The same tetramer compositions at  $\epsilon \in \{-5, 0, +5\}\%$  yield  $\pi$ -projected DOS, HOMO–LUMO gaps, and band-edge energies.

**Polaron localization ( $2 \times 2$  factorial, four calculations).** IPR and nearest-neighbor coupling  $J$  are evaluated on a  $2 \times 2$  grid: pristine vs. B doping crossed with  $\epsilon \in \{0, +3\}\%$  (converged neutral single-points; Supplemental Material Fig. S6).

**Charged polarons (Supplemental Material).** Tetramer GEO.OPT at  $q = \pm 1$  and vertical single-points at neutral geometry target Marcus  $\lambda^\pm$  [1]; they are not used for  $\mathcal{S}$  or  $\alpha$  in the main text.

**Non-additive size scaling (40 calculations).** Periodic cells at  $\epsilon \in \{0, +3\}\%$  for each dopant and  $n = 1-8$  supply  $\mathcal{S}(n)$ ; dopant count is one site per  $C_{60}$  unit (Table S2).

**Reference geometries (Supplemental Material).** Neutral tetramer relaxations and a pristine single-point on the relaxed cell define charged-state input geometries.

### Energy observables

Linear strain sensitivity  $\alpha = dE/d\epsilon$  (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive  $\alpha$  means tension *raises* total energy; negative  $\alpha$  means tension *lowers* it under the rigid-strain protocol. Substitution energy  $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates  $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$ ; the neglected term is precisely  $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at  $n = 4$  and  $+3\%$  strain, P has  $|\mathcal{S}| \approx 24$  meV/atom while the per-atom substitution scale is  $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$  eV/atom ( $|E_f| \approx 16$  eV per P site,  $n_{\text{dop}}=12$ , Table S1)—i.e.  $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$ , large enough to reorder stability margins in sequential doping-then-strain screens even though  $|\mathcal{S}|$  exceeds the rigid per-atom strain increment ( $\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$  meV/atom for P).

**Non-additive coupling metric.** We quantify strain–doping cross terms with the synergy order parameter  $\mathcal{S}$ :

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where  $\epsilon$  is biaxial strain,  $\delta \in \{B, N, P\}$ , and  $\Delta E(\epsilon, \delta)$  is the energy per atom relative to the pristine, zero-strain reference at the same supercell size  $n \times C_{60}$ . The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives  $\mathcal{S} = 0$ ;  $\mathcal{S} < 0$  indicates cooperative stabilization beyond the two-scan sum, whereas  $\mathcal{S} > 0$  indicates cooperative destabilization.

To first order in strain,  $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$  when  $\alpha_\delta = dE/d\epsilon$  is evaluated for the doped cell and  $\alpha_0$  for the pristine reference at the same  $n$  (rigid-strain protocol); this links the mechanical coefficient  $\alpha$  to the cross term at small  $\epsilon$ .

Periodic-cell  $\mathcal{S}$  values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report  $\mathcal{S}$  for total energy only; an analogous gap cross term  $\mathcal{S}_{\text{gap}}$  is not computed here—Fig. 1(a) shows that gap slopes need not track  $|\mathcal{S}|$ .

**Localization and coupling (Exp. 4).** Following Capobianco *et al.* [1], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling  $J$  from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8),  $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ , on a Mulliken-projected orbital grid;  $J$  is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables

are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant  $J \simeq 75$ –81 meV reported for pristine qHP networks [1], but they provide an internally consistent trend under coupled  $(\epsilon, \delta)$ .

**Polaron binding (Supplemental Material).** Marcus reorganization energies follow the vertical-adiabatic decomposition [1]:  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ , with separate internal ( $\lambda_{\text{loc}}^\pm$ ) and environmental ( $\lambda_{\text{env}}^\pm$ ) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

## RESULTS

Results are organized parallel to Ref. 1: band-edge electronics and localization precede transport-relevant coupling, after which we report mechanical and non-additive energy cross terms specific to coupled  $(\epsilon, \delta)$  tuning.

### Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at  $\epsilon \in \{-5, 0, +5\}\%$ . B introduces hole states near the valence manifold (acceptor character), compressing the  $\pi$  gap toward zero at  $\epsilon = 0$ , whereas N donates electrons into the conduction manifold (donor character) and P (larger covalent radius) perturbs the network more heterogeneously—hence the dopant-specific gap slopes in panel (a). Figure 1(b) shows  $\pi$ -DOS for B @  $\epsilon = 0$ .

### Polaron localization and electronic coupling

Capobianco *et al.* [1] established that qHP electron polarons remain molecularly localized while  $J$  grows with shorter interfullerene distances. A  $2 \times 2$  factorial separates strain-only, doping-only, and coupled contributions (Supplemental Material Fig. S6): pristine @  $\epsilon=0$  (IPR = 75.0,  $J=26.6$  meV); pristine @ +3% (IPR = 76.1,  $J=26.5$  meV); B @  $\epsilon=0$  (IPR = 24.6,  $J=46.4$  meV); B @ +3% (IPR = 23.1,  $J=37.2$  meV). B substitution accounts for most of the IPR drop ( $\Delta\text{IPR} \approx -50$ ) and the zero-strain  $J$  enhancement ( $\Delta J \approx +20$  meV); +3% strain on pristine is negligible, but under B doping it *lowers*  $J$  by  $\sim 9$  meV (negative  $J$  cross term). This transport-level non-additivity arises from orbital-coupling sensitivity to interfullerene spacing and is *not* numerically equivalent to total-energy  $\mathcal{S}$  (mechanical/electrostatic cross terms in  $E_{\text{tot}}$ ). Using literature  $\lambda \sim 0.1$  eV as an order-of-magnitude bound [1], all four points satisfy  $J \ll \lambda/2$

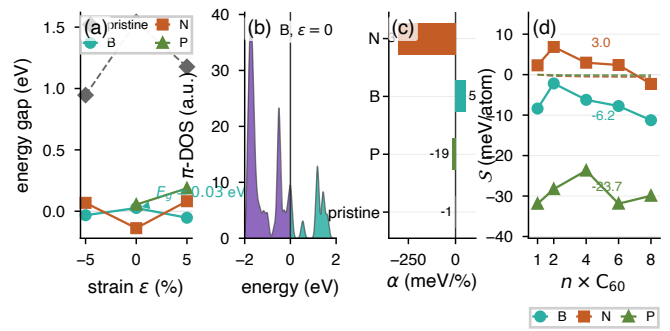


FIG. 1. Coupled  $(\epsilon, \delta)$  response in doped qHP  $C_{60}$  (PBE+D3). (a) HOMO–LUMO gap vs. strain. (b)  $\pi$ -DOS for B @  $\epsilon=0$  (Electron SI palette). (c) Mechanical  $\alpha$ . (d)  $\mathcal{S}(n)$  at +3%; inset: polaron IPR/ $J$  factorial.

(small-polaron hopping); this regime is inferred, not measured here. Figure 1(d) inset and Supplemental Material Fig. S6 summarize the grid; absolute  $J$  is below pristine-network values because of fragment size and functional choice.

### Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ( $|\alpha| \approx 1$  meV/%). Substitution opens a wide spread: N ( $\alpha = -303.2$  meV/%, tension stabilizes), B (+56.9 meV/%, tension destabilizes), P (−19.1 meV/%) (Table S1). Figure 1(c) highlights the opposite  $\alpha$  for N vs. B; ranking dopants by  $\epsilon = 0$  stability alone is misleading.

### Non-additive strain–doping coupling

Figure 1(d) tracks  $\mathcal{S}(n)$  with one dopant per  $C_{60}$ . At  $n = 4$  and +3% strain,  $\mathcal{S} = -6.2$  meV/atom (B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude. Fifteen coupled points span  $n \in \{1, 2, 4, 6, 8\}$  and B/N/P at +3%. For N,  $\mathcal{S}$  changes sign between  $n = 6$  (+2.4 meV/atom) and  $n = 8$  (−2.3 meV/atom). Because  $n = 6$ –8 use 280 Ry whereas  $n \leq 5$  use 300 Ry, this reversal may partly reflect basis-set change rather than intrinsic size physics; we treat it as provisional pending a matched-cutoff  $n = 6$  benchmark.

## DISCUSSION

**Relation to Ref. 1.** Capobianco *et al.* [1] compared *pristine* vdW and qHP  $C_{60}$  and identified enhanced  $J$ —not reduced  $\lambda$ —as the dominant mobility lever, with polaron hopping retained in both phases. Our study transfers that observables vocabulary (gap edges, IPR,  $J$ ) to *doped* tetramers under strain and adds an en-

ergy cross term  $\mathcal{S}$  absent from sequential doping-only or strain-only scans [3, 4]. Where Khan and Leuenberger [4] classify impurity-engineered *half-semiconductor* gaps at fixed geometry, our  $\mathcal{S}$  analysis tests total-energy additivity under identical biaxial strain—complementary metrics for impurity–strain co-design [6]. The factorial  $J$  pattern (B raises  $J$  at  $\epsilon=0$  but +3% strain partially reverses it) is qualitatively aligned with Capobianco’s distance–coupling picture: B acceptor states enhance interfullerene hybridization, making  $J$  more distance-sensitive, so tensile strain that increases spacing suppresses  $J$  far more under B doping than in pristine cells. Transport and total-energy non-additivity are related in origin but not one-to-one;  $\mathcal{S}$  should not be used to infer hopping-rate changes without independent  $J/\lambda$  data.

**Mechanistic synthesis.** (i) **Opposite  $\alpha$  for N vs. B.** N introduces excess  $\pi$  electrons that strengthen bonding under tension (negative  $\alpha$ ), whereas B creates electron deficiency and antibonding  $\pi^*$  character that rises with lattice expansion (positive  $\alpha$ )—consistent with donor vs. acceptor  $\pi$ -counting and the PDOS shifts in Fig. 1(a,b). (ii) **Large  $|\mathcal{S}|$  for P despite small  $|\alpha|$ .** P’s covalent radius ( $\sim 107$  pm vs.  $\sim 77$  pm for C) and longer P–C bonds impose intrinsic cage distortion; rigid external strain couples to this built-in local stress, yielding a cross term that is invisible in separate scans but large when strain and substitutional doping are applied jointly—consistent with P’s intermediate  $E_f$  yet largest  $|\mathcal{S}|$  at +3%. (iii) **N  $\mathcal{S}$  sign change with  $n$ .** With one dopant per  $C_{60}$ , electrostatic and strain fields are dilute at large  $n$ ; the sign flip between  $n = 6$  and  $n = 8$  may reflect finite-size electrostatics but is confounded with the 280 Ry cutoff introduced at  $n \geq 6$ —interpretation is deferred until a constant-cutoff control.

(iv) **Linear  $\mathcal{S}$ – $\alpha$  check.** Using tetramer  $\alpha$  (Table S1) in  $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$  at  $\epsilon = +3\%$  and dividing by 240 sites gives  $\sim +0.7$ ,  $-3.8$ , and  $-0.2$  meV/atom for B/N/P—far smaller than the periodic  $n=4$  values ( $-6.2$ ,  $+3.0$ ,  $-23.7$  meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic  $\mathcal{S}$ ; for P, nonlinear coupling dominates beyond first-order  $(\alpha_\delta - \alpha_0)\epsilon$ .

**Design implications.** At +3% strain, P carries the largest additive-approximation error in  $\mathcal{S}$  ( $|\mathcal{S}| \sim 24$ – $32$  meV/atom); co-design should treat P under tension as a *coupled* stability problem, not  $|E_f|$  at  $\epsilon = 0$  alone. Where  $\mathcal{S} < 0$  (B and P at  $n=4$ ), coupled tension can stabilize doped cells beyond additive predictions; where  $\mathcal{S} > 0$  (N at  $n=4$ ), sequential screens overestimate stability under load. Gap closing [Fig. 1(a)] does not track  $|\mathcal{S}|$ ; we do not compute  $\mathcal{S}_{\text{gap}}$ , so photonic or transport targets cannot be inferred from total-energy  $\mathcal{S}$  alone. N  $\mathcal{S}$  sign reversal by  $n = 8$  cautions against tetramer-only extrapolation.

**Limitations and outlook.** Rigid strain (upper-

bound on  $\mathcal{S}$ ); fixed-seed substitutional sites without thermodynamic ranking; mixed 280/300 Ry cutoffs across  $n$  (N  $\mathcal{S}$  sign change at  $n = 6$ – $8$ ); tetramer vs. scaling concentration mismatch; PBE+D3 without Koopmans/rVV10; gap non-additivity ( $\mathcal{S}_{\text{gap}}$ ) not quantified; Marcus  $\lambda$  and vertical transitions from charged geometry optimization still pending (Supplemental Material).

## CONCLUSIONS

We quantified non-additive strain–doping coupling in heteroatom-substituted qHP  $C_{60}$ . Opposite  $\alpha$  for N vs. B and finite  $\mathcal{S}$  up to  $\sim 32$  meV/atom show that joint  $(\epsilon, \delta)$  tuning is not equivalent to sequential strain and doping scans; for P under tension, the cross term is a few percent of the per-atom substitution-energy scale yet can significantly alter stability rankings in sequential screening workflows. These energy-landscape metrics complement—but do not replace—Marcus  $\lambda$  from charged-state vertical transitions needed to connect  $\mathcal{S}$  to polaron hopping rates.

We acknowledge computational resources and support from Xiamen University.

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